

Reproducing the CHC Western-Indian-Ocean / RONI Outlook for OND 2026

A worked walk-through of the CHC / FEWS NET companion deck — "IOD, Western Indian Ocean & RONI outlook for June 2026" (Laura Harrison & Chris Funk) — rebuilt with the `rosetta` and `deepscale` libraries.

What the deck argues

East Africa's **October–December (OND) "short rains"** are strongly controlled by the state of the two neighbouring oceans. The deck's case for 2026 is that the ocean is moving toward the configuration that most reliably produces *extreme* short rains: a **strong El Niño** in the Pacific, a **strong positive Indian Ocean Dipole (IOD)**, and an **exceptionally warm western Indian Ocean pole** (the deck's slide 3: "*expected... very warm (>29°C)*") that supplies moisture onshore. The deck's headline comparison is that OND 2026 resembles **1997 and 2023** (slide 2), with **2015 and 2019** in a broader set of warm-western-pole analog years.

What this notebook builds — and on which data

The workflow uses only `rosetta` for data acquisition and `deepscale` for the analysis. It goes figure by figure, in the deck's slide order.

Two fidelity notes. (1) The deck averages **six** NMME models; here we use the three with a full 1991–2020 hindcast and a live June-2026 OND forecast available through `rosetta` — CanSIPS-IC4, CCSM4, and CESM1. Two of the three (CCSM4, CESM1) share NCAR/CESM lineage, so the effective ensemble diversity is a little lower than "three models" suggests. (2) ERSSTv5 is the observed SST, as in the deck.

Setup

```
import warnings; warnings.filterwarnings("ignore")
import numpy as np, pandas as pd, xarray as xr
import matplotlib.pyplot as plt
import rosetta
import deepscale as ds
import deck_style as deck           # CHC palettes / marker styles, sampled from the deck legends

BASELINE = (1991, 2020)             # the deck's climatological reference period
YEARS = list(range(1991, 2026))     # the hindcast record the forecast is scored against
TARGET = 2026                       # the season being forecast
EEA = [-5, 12, 36, 52]              # eastern East Africa box (Kenya, Somalia, S/E Ethiopia) for
rainfall
```

1 • The ocean data — a single `fetch`

The observed record is **ERSSTv5**, the series CHC uses, fetched over OPeNDAP in one call (Rosetta also guards NOAA's silent zero-fill-on-over-large-request failure mode). We reduce SST to the **October–December** seasonal mean per year:

```
sst = rosetta.fetch("obs/ersst-v5", "sst", hindcast=(1991, 2026), verbose=False)["sst"]
ond = sst.sel(time=sst.time.dt.month.isin([10, 11, 12])).groupby("time.year").mean("time")
print("OND-mean SST, one field per year:", dict(ond.sizes))
```

```
OND-mean SST, one field per year: {'year': 35, 'lat': 89, 'lon': 180}
```

2 · Reducing SST to three indices

Three scalar indices per year, each a DeepScale **Index** (which boxes, how each is transformed, how they combine). RONI is catalogued; the dipole and the western pole we define explicitly:

```
roni = ds.Index.named("roni", baseline=BASELINE).reduce(ond)      # relative Oceanic Niño index

# Indian Ocean Dipole (Saji et al. 1999): west pole minus east pole, each an SST anomaly (°C).
dmi_index = ds.Index.custom(
    name="dmi",
    regions={"west": [-10, 10, 50, 70], "east": [-10, 0, 90, 110]},
    combine=lambda box: box["west"] - box["east"],
    transform="anomaly", weights="cos_lat", baseline=BASELINE)

# Western Indian Ocean pole in ABSOLUTE °C, for the deck's ~29 °C threshold.
wio_index = ds.Index.custom(
    name="wio", regions={"west": [-10, 10, 50, 70]},
    combine=lambda box: box["west"], transform="raw", weights="cos_lat")

iod = dmi_index.reduce(ond)
wio = wio_index.reduce(ond)

table = pd.DataFrame({"RONI (°C)": roni.to_series(), "IOD (°C)": iod.to_series(),
                      "WIO (°C)": wio.to_series()}).round(2)
table.loc[[1997, 2015, 2019, 2023]]
```

	RONI (°C)	IOD (°C)	WIO (°C)
YEAR			
1997	1.93	1.14	28.639999
2015	1.93	0.35	29.120001
2019	0.21	1.23	28.920000
2023	1.23	1.25	29.190001

The table tracks the deck's reasoning — 1997 strong on RONI, 2019 a strong dipole with a modest El Niño, all four warm on WIO — **with one flag the deck's own numbers carry: 2015's IOD is weak** (a decaying dipole), so 2015 qualifies as an analog only through the absolute WIO temperature, not the dipole. We return to that in the final section; it is the deck's choice, faithfully reproduced here.

2b · The rainfall outcomes — restoring the link the figure is *about*

The deck's flagship scatter is not a plot of ocean states; it is a plot of ocean states **coloured by the East-African short-rains outcome each one produced** (deck slide 2: *"The colors of points indicate the rainfall tercile outcomes for the eastern East Africa region, based on CHIRPS3 data and a 1991-2020 baseline"*). That colouring is the entire scientific claim — "the upper-right ocean state is where the wet extremes fall." We compute it from CHIRPS over

the eastern-East-Africa box, as an OND regional-mean per year, categorised into terciles against the 1991–2020 climatology:

```
rain = rosetta.fetch("obs/chirps-v2-monthly", "precip", target="OND", seasonal="mean",
                    region=EEA, hindcast=(1991, 2025), allow_partial=True, verbose=False)
["precip"]
rain = rain.mean(["lat", "lon"]).dropna("year")  # EEA regional-mean OND rate per
year

clim = rain.sel(year=slice(BASELINE[0], BASELINE[1]))
t33, t67 = float(clim.quantile(1/3)), float(clim.quantile(2/3))
rain_tercile = xr.where(rain < t33, "Below", xr.where(rain > t67, "Above", "Normal"))
rain_tercile.name = "EEA OND rainfall tercile"

print(f"EEA OND rainfall terciles (1991–2020 baseline: below<{t33:.2f}, above>{t67:.2f} mm/day)")
for cat in ["Above", "Normal", "Below"]:
    yrs = [int(y) for y, c in zip(rain.year.values, rain_tercile.values) if c == cat]
    print(f" {cat:6s}: {yrs}")
```

```
EEA OND rainfall terciles (1991–2020 baseline: below<1.27, above>1.70 mm/day)
Above : [1994, 1997, 2000, 2002, 2006, 2009, 2011, 2013, 2015, 2019, 2023]
Normal: [1992, 1995, 1999, 2001, 2004, 2008, 2012, 2014, 2017, 2018, 2022, 2024]
Below : [1991, 1993, 1996, 1998, 2003, 2005, 2007, 2010, 2016, 2020, 2021, 2025]
```

This is the observed rainfall context. Note where the deck's four analog years land: they should be the **Above**-tercile (wet) years if the ocean-state argument holds — that is the relationship the scatter is meant to display.

3 · From model output to a calibrated forecast

The deck's *forecast* for OND 2026 comes from the **NMME**, run from a June start. We fetch each model's OND SST forecast, average across models, reduce to the same three indices, and calibrate: **quantile_map** onto the observed distribution, **error_bounds** for the 80% interval. This reproduces the deck's method exactly — including two properties the final section revisits: the quantile map here uses the deck-default **extrapolate="clamp"**, and the interval is built from in-sample 1991–2020 forecast errors.

```

MODELS = ["nmme/cansipsic4", "nmme/ccsm4", "nmme/cesm1"]

def nmme_ond_sst(year):
    fields = []
    for m in MODELS:
        d = rosetta.fetch(m, "sst", init=f"{year}-06", target="OND", region=[-25, 25, 0, 360],
                          hindcast=(year, year), verbose=False)["sst"] # tropical band covers
all index boxes
        if d.sizes.get("init_time", 0) == 0: # provider gap for this model-year ->
skip
            continue
        f = d.isel(init_time=0).mean("member")
        if float(np.nanmax(np.abs(f))) > 0: # drop an all-zero (NMME zero-fill) field
            fields.append(f)
    return xr.concat(xr.align(*fields, join="inner"), dim="model").mean("model") # mean over
available models

fc_sst = xr.concat([nmme_ond_sst(y) for y in YEARS + [TARGET]],
                    dim=pd.Index(YEARS + [TARGET], name="year"))

def calibrate(index_obj, observed, extrapolate="clamp"):
    fc = index_obj.reduce(fc_sst)
    hind = list(range(BASELINE[0], BASELINE[1] + 1))
    fc_h, obs_h = fc.sel(year=hind), observed.sel(year=hind)
    corrected = ds.quantile_map(fc, fc_h, obs_h, extrapolate=extrapolate)
    corrected = xr.DataArray(corrected, dims="year", coords={"year": fc.year})
    bounds = ds.error_bounds(corrected.sel(year=hind), obs_h,
                              float(corrected.sel(year=TARGET)), level=0.8)
    return corrected, bounds

roni_fc, roni_b = calibrate(ds.Index.named("roni", baseline=BASELINE), roni)
iod_fc, iod_b = calibrate(dmi_index, iod)
wio_fc, wio_b = calibrate(wio_index, wio)

for name, fc, b in [("RONI", roni_fc, roni_b), ("IOD", iod_fc, iod_b), ("WIO", wio_fc, wio_b)]:
    v = float(fc.sel(year=TARGET))
    print(f" 2026 {name:4s} forecast = {v:+6.2f} °C [80% CI {float(b.lower):+.2f},
{float(b.upper):+.2f}]"

```

```

2026 RONI forecast = +1.93 °C [80% CI +1.21, +2.69]
2026 IOD forecast = +1.23 °C [80% CI +0.85, +1.66]
2026 WIO forecast = +29.12 °C [80% CI +28.89, +29.34]

```

4 · The figures

```
def obs_vs_forecast(observed, forecast, title, ylabel):
    fig, ax = plt.subplots(figsize=(9, 3.6))
    yrs = observed.year.values
    ax.plot(yrs, observed.sel(year=yrs), color="#1f77b4", lw=1.8, marker="o", ms=3,
label="Observed (ERSSTv5)")
    ax.plot(forecast.year.values, forecast, color="#111", lw=1.4, marker="o", ms=3,
label="Forecast (NMME, bias-corrected)")
    ax.axvspan(BASELINE[0], BASELINE[1], color="#000", alpha=0.05, zorder=0) # training window
shaded
    ax.scatter([TARGET], [float(forecast.sel(year=TARGET))], color="#d62728", s=90, zorder=5,
label="2026")
    ax.set_title(title, fontweight="bold"); ax.set_ylabel(ylabel)
    ax.legend(fontsize=8, ncol=3, frameon=False); ax.grid(alpha=0.3)
    fig.tight_layout(); plt.show()
```

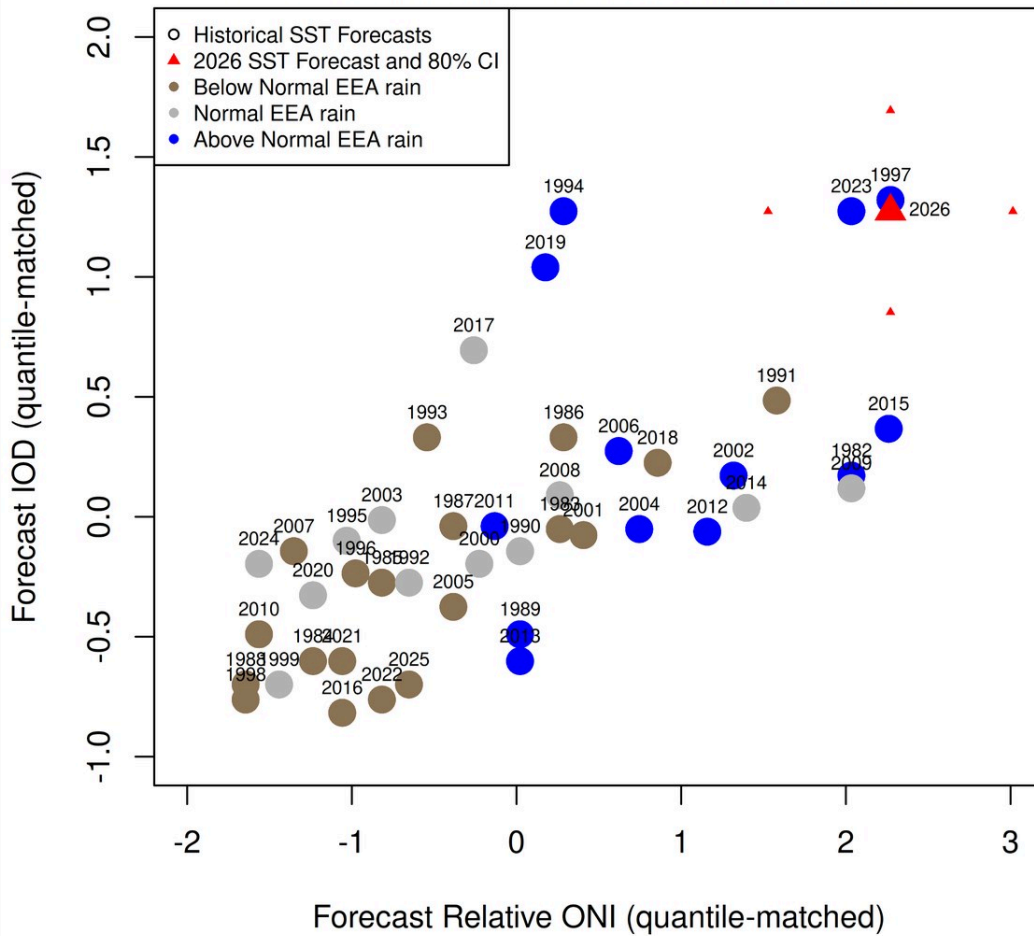
Slide 2 · RONI vs IOD, coloured by rainfall outcome

The deck's summary slide places every historical year on a **forecast** RONI (x) vs **forecast** IOD (y) plane — the quantile-matched forecasts, not observations — colours each point by the observed eastern-East-Africa OND rainfall tercile it produced, and marks the 2026 forecast with 80% whiskers.

OND SST Forecasts: ENSO and Indian Ocean Dipole

Forecast month: June

Outlook: NMME Ensemble Mean



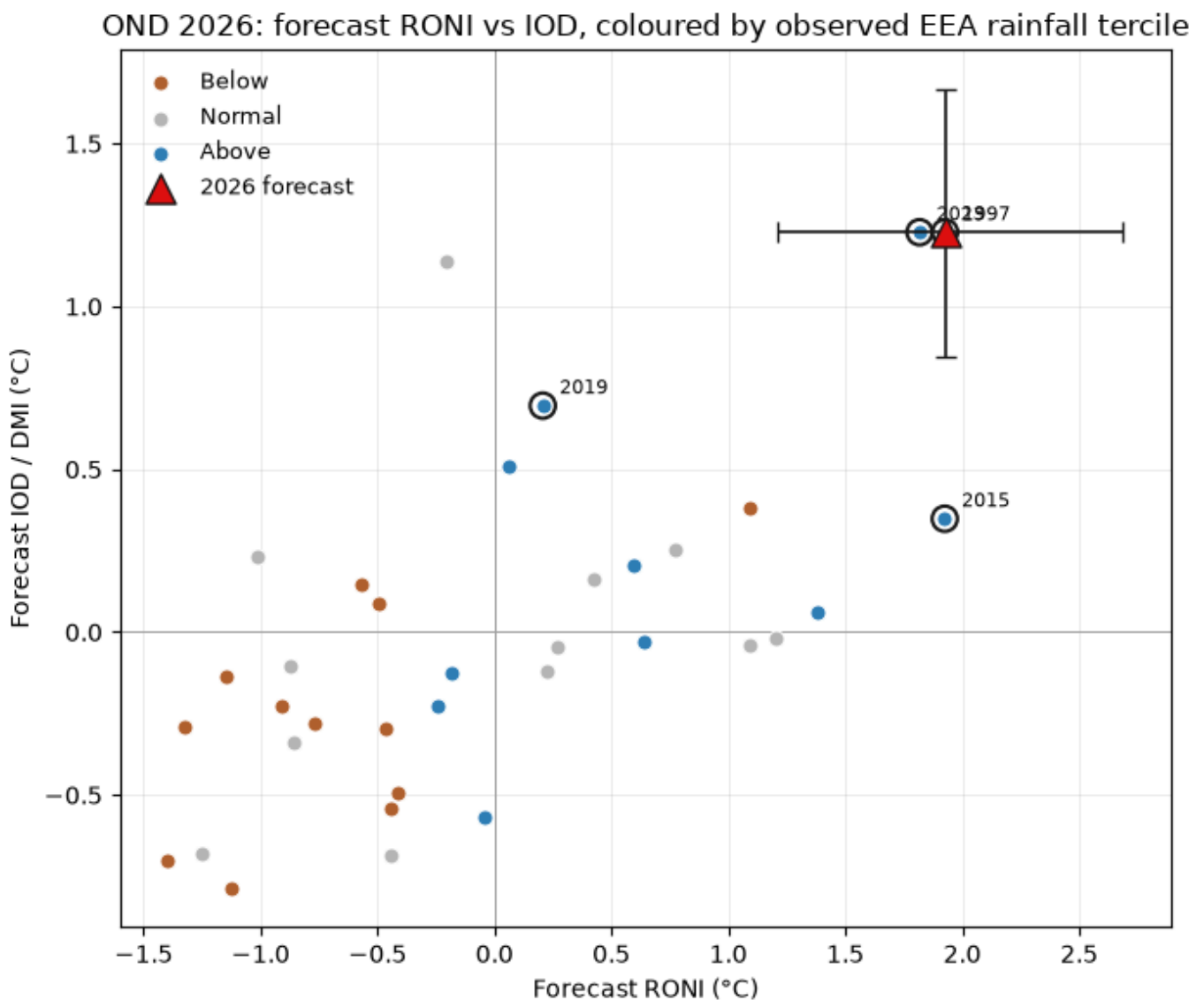
CHC deck · slide 2 — OND SST Forecasts: ENSO and Indian Ocean Dipole (RONI vs IOD, June forecast)

The axes are the **quantile-matched forecasts** `roni_fc` / `iod_fc` (as the deck's are), and the points are coloured by the **observed rainfall tercile** computed in §2b.

```

hist_yrs = rain_tercile.year                                # historical years with an
observed rainfall outcome
fig = ds.plot_index_scatter(
    roni_fc.sel(year=hist_yrs), iod_fc.sel(year=hist_yrs),    # FORECAST axes (deck), 2026
    drawn_as_the_triangle
    color_by=rain_tercile, categories=["Below", "Normal", "Above"],
    colors=["#b2622d", "#b8b8b8", "#2d7db8"],                # dry / normal / wet
    highlight=[1997, 2015, 2019, 2023],
    forecast=(float(roni_fc.sel(year=TARGET)), float(iod_fc.sel(year=TARGET))),
    error_bars=(roni_b, iod_b),
    forecast_marker="^", forecast_color="#e01010", forecast_label="2026 forecast",
    xlabel="Forecast RONI (°C)", ylabel="Forecast IOD / DMI (°C)",
    title="OND 2026: forecast RONI vs IOD, coloured by observed EEA rainfall tercile")
plt.show()

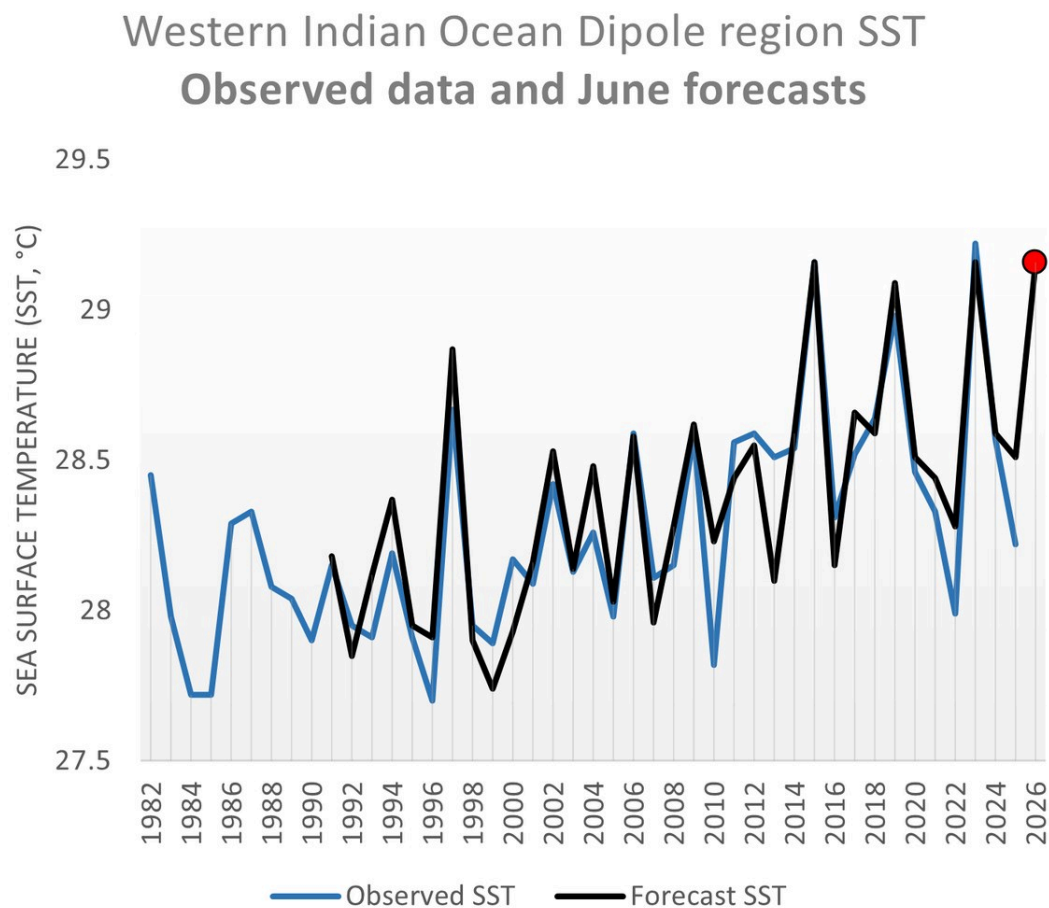
```



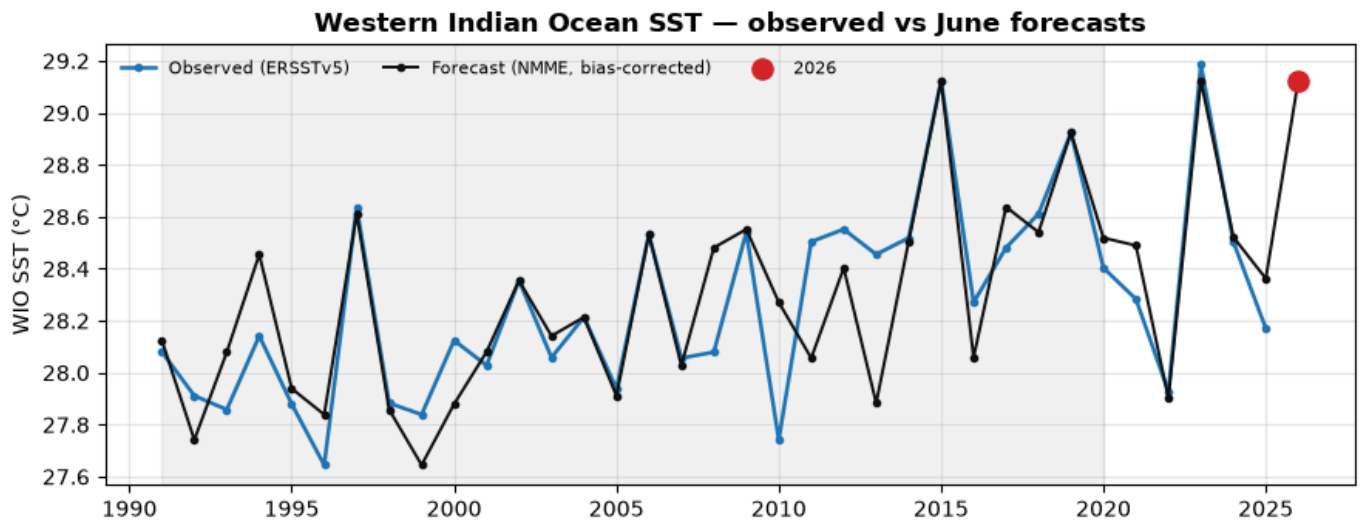
Now the figure makes the deck's actual claim: the **wet (blue) years cluster in the upper-right**, and the 2026 forecast falls among them — closest to **1997 and 2023** (the deck's headline pair). The colouring also makes visible what the index table flagged: **2015 sits low on the IOD axis**, an outlier among the four circled analogs in the very dimension the argument privileges — so its inclusion rests on the western-pole temperature alone.

Slide 3 · Western Indian Ocean SST — observed vs forecast

The forecast rises to near the top of the record. That the metric is an *absolute* temperature on a warming trend is the deck's choice — examined in the final section.




```
obs_vs_forecast(wio, wio_fc, "Western Indian Ocean SST – observed vs June forecasts", "WIO SST (°C)")
print(f"2026 WIO forecast: {float(wio_fc.sel(year=TARGET)):.2f} °C (deck threshold ~29 °C)")
```

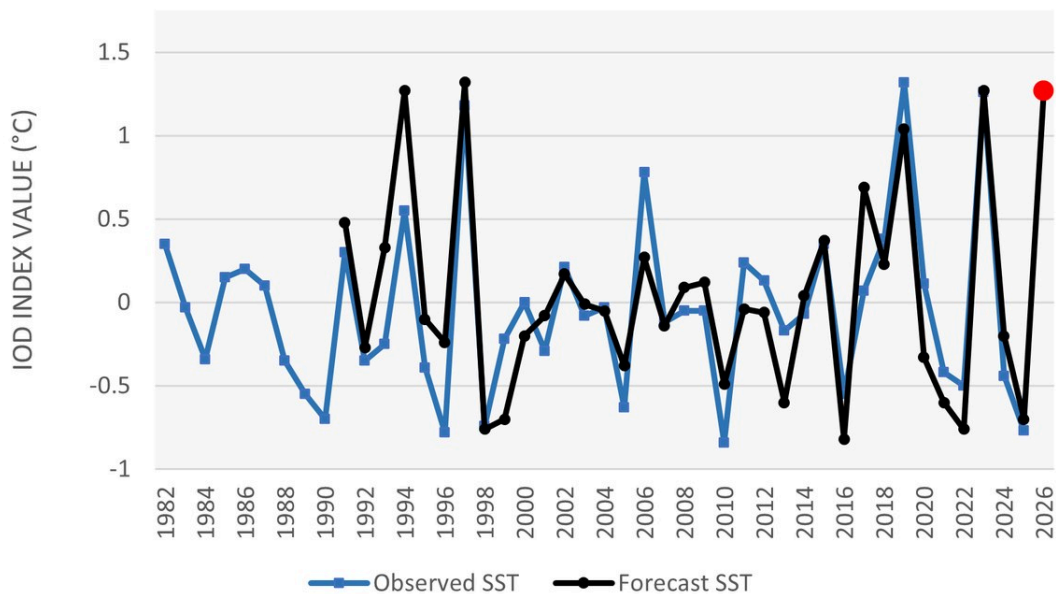


2026 WIO forecast: 29.12 °C (deck threshold ~29 °C)

Slide 4 · Indian Ocean Dipole — observed vs forecast

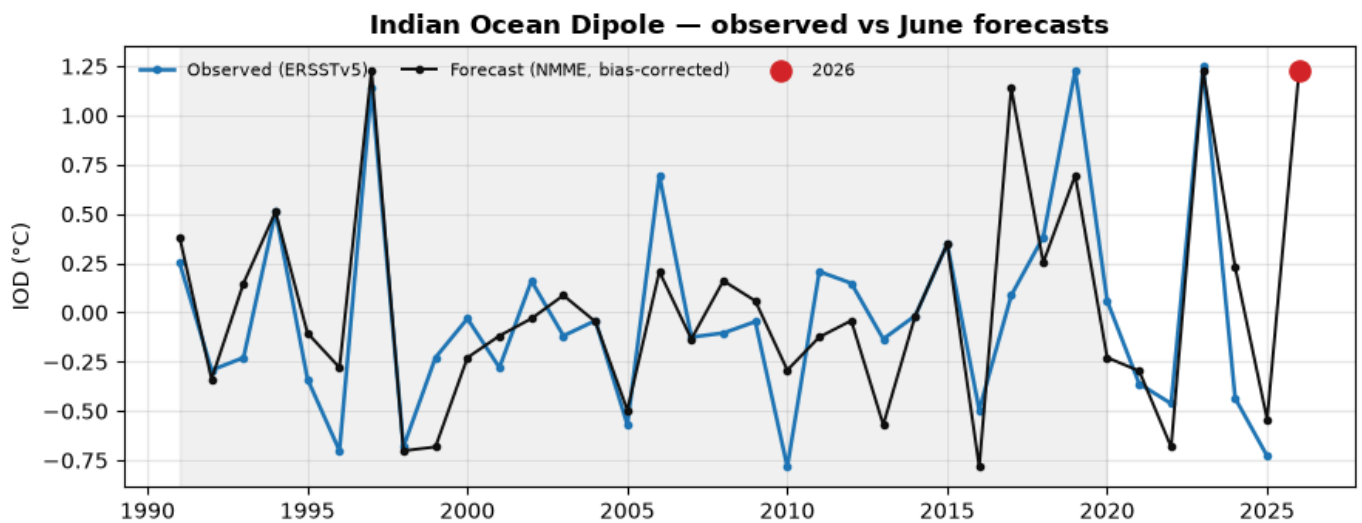
2026 sits among the strongest positive-dipole years. The confidence of a *June-initialised* OND-IOD forecast is the subject of the final section.

Indian Ocean Dipole Observed data and June forecasts



CHC deck · slide 4 — Indian Ocean Dipole — observed data and June forecasts

```
obs_vs_forecast(iod, iod_fc, "Indian Ocean Dipole – observed vs June forecasts", "IOD (°C)")
print(f"2026 IOD forecast: {float(iod_fc.sel(year=TARGET)):+.2f} °C")
```

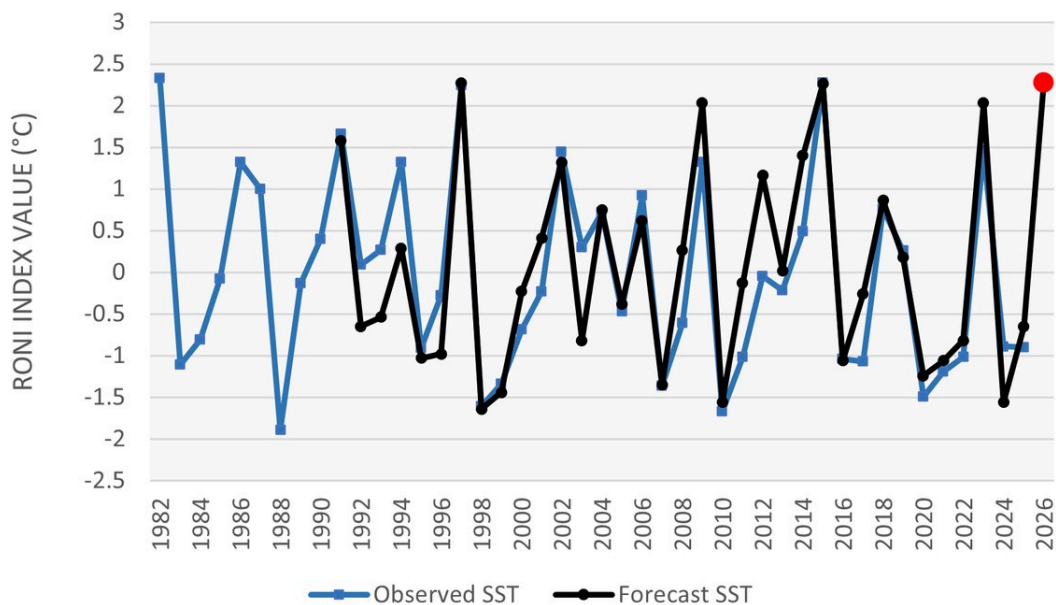


2026 IOD forecast: +1.23 °C

Slide 5 · Relative Oceanic Niño Index — observed vs forecast

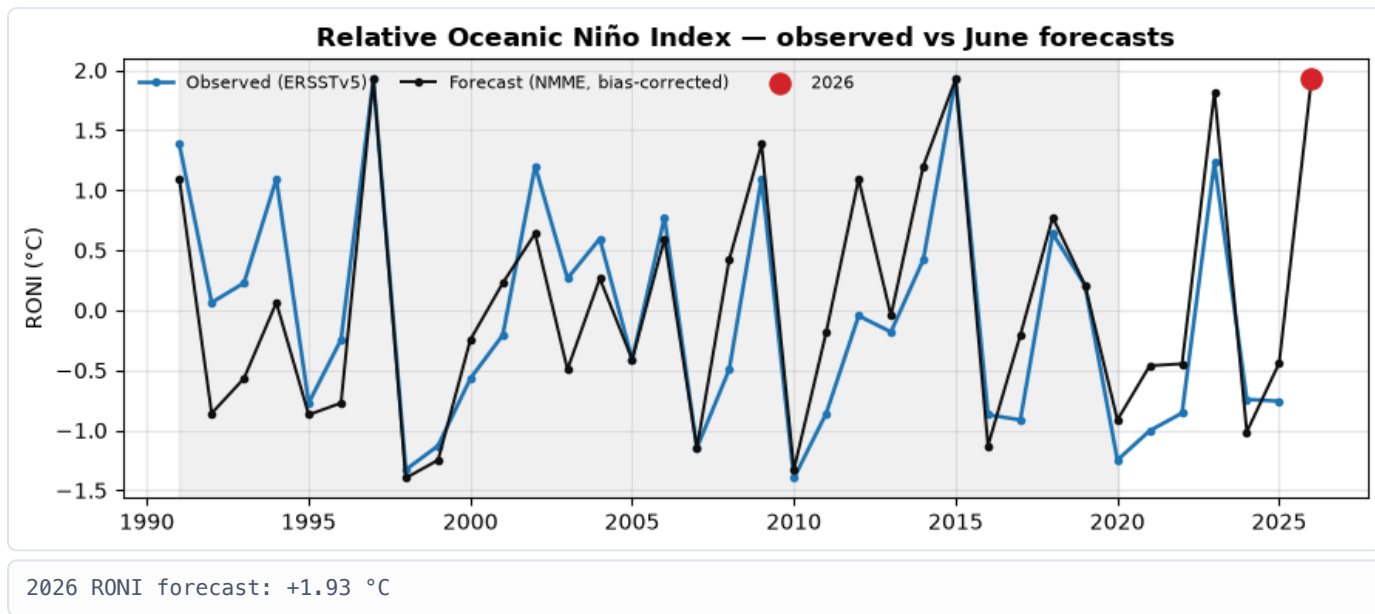
2026 is forecast as one of the strongest RONI years. Note the value sits *exactly* at the historical maximum — a fingerprint of the clamped quantile map, taken up in the final section.

Relative Oceanic Nino Index Observed data and June forecasts



CHC deck · slide 5 — Relative Oceanic Niño Index — observed data and June forecasts

```
obs_vs_forecast(roni, roni_fc, "Relative Oceanic Niño Index – observed vs June forecasts", "RONI (°C)")
print(f"2026 RONI forecast: {float(roni_fc.sel(year=TARGET))+.2f} °C")
```



5 · The analog years — the deck's rule, reproduced (descriptively)

The deck names **1997, 2015, 2019, 2023**. A simple threshold rule recovers that list; since the thresholds are chosen to match the published set, it is a *description* of the deck's analogs rather than an independent test. It is worth showing how sharp the selection is:

```
for thr in [1.0, 1.1, 1.2]:
    u = ds.analogs_where(iod > thr) | ds.analogs_where(wio > 29.0)
    print(f" IOD>{thr:.1f} | WIO>29.0 -> {sorted(u.years.tolist())}")
print(" deck's stated set -> [1997, 2015, 2019, 2023]")
```

```
IOD>1.0 | WIO>29.0 -> [1997, 2015, 2019, 2023]
IOD>1.1 | WIO>29.0 -> [1997, 2015, 2019, 2023]
IOD>1.2 | WIO>29.0 -> [2015, 2019, 2023]
deck's stated set -> [1997, 2015, 2019, 2023]
```

The set is not sharp: a slightly looser dipole threshold pulls in more years. The deck's four are a reasonable *description* of the strong-signal end of the record, not a bright line — and 2015 depends entirely on the WIO clause.

6 · Beyond the deck — four methodological notes

The reproduction above follows the deck's method. This short section revisits four choices in that method and shows how alternatives change the numbers: the clamped quantile map, the in-sample error bounds, the absolute WIO threshold, and the June-initialised IOD forecast skill.

6a · Clamped vs extrapolated quantile mapping

`quantile_map` defaults to `extrapolate="clamp"`, which caps a forecast beyond the training range at the strongest event in that range. A strong-El-Niño June forecast exceeds the 1991–2020 hindcast maximum, so the clamped RONI lands on the observed record; `extrapolate="linear"` continues the transfer function past it:

```
roni_lin, roni_lb = calibrate(ds.Index.named("roni", baseline=BASELINE), roni,
extrapolate="linear")
iod_lin, iod_lb = calibrate(dmi_index, iod, extrapolate="linear")
wio_lin, wio_lb = calibrate(wio_index, wio, extrapolate="linear")

print(f"{'index':5s} {'clamped':>10s} {'linear':>10s}    (2026 forecast)")
for name, c, l in [("RONI", roni_fc, roni_lin), ("IOD", iod_fc, iod_lin), ("WIO", wio_fc,
wio_lin)]:
    print(f"{'name':5s} {float(c.sel(year=TARGET)):>+10.2f} {float(l.sel(year=TARGET)):>+10.2f}")
```

index	clamped	linear	(2026 forecast)
RONI	+1.93	+1.93	
IOD	+1.23	+1.70	
WIO	+29.12	+32.70	

Under clamping the RONI and IOD forecasts sit at the historical maximum; linear extrapolation lets 2026 exceed it. So "comparable to the strongest years" partly reflects the clamp rather than the raw forecast. For WIO — an absolute temperature on a warming trend — clamping to the 1991–2020 maximum is a mild low bias, since some post-2020 observed years already exceed it.

6b · In-sample vs cross-validated error bounds

The 80% interval uses residuals of the quantile-matched hindcast over the same 1991–2020 used to fit the map, which empirical quantile mapping shrinks by construction. A leave-one-year-out (LOYO) interval — refit without year k , predict k — gives an out-of-sample estimate:

```
def loyo_halfwidth(index_obj, observed, level=0.8):
    fc = index_obj.reduce(fc_sst); hind = list(range(BASELINE[0], BASELINE[1] + 1))
    errs = []
    for yr in hind:
        tr = [y for y in hind if y != yr]
        pred = ds.quantile_map(fc.sel(year=[yr]), fc.sel(year=tr), observed.sel(year=tr),
extrapolate="linear")
        errs.append(float(np.asarray(pred).ravel()[0]) - float(observed.sel(year=yr)))
    lo, hi = np.quantile(errs, [(1 - level) / 2, 1 - (1 - level) / 2])
    return (hi - lo) / 2

for name, idx, obs, b in [("RONI", ds.Index.named("roni", baseline=BASELINE), roni, roni_b),
                          ("IOD", dmi_index, iod, iod_b), ("WIO", wio_index, wio, wio_b)]:
    insample = (float(b.upper) - float(b.lower)) / 2
    print(f" {name:4s} in-sample ±{insample:.2f} °C    ->    LOYO ±{loyo_halfwidth(idx, obs):.2f} °C")
```

RONI	in-sample ±0.74 °C	->	LOYO ±0.84 °C
IOD	in-sample ±0.41 °C	->	LOYO ±0.45 °C
WIO	in-sample ±0.22 °C	->	LOYO ±0.30 °C

The cross-validated intervals are somewhat wider, so the in-sample bounds are a lower estimate of the June-initialised forecast uncertainty.

6c · Absolute vs trend-relative western-pole warmth

The WIO index is a raw absolute temperature; on a basin that has warmed $\sim 1^\circ\text{C}$ since the 1980s, a fixed 29°C cut increasingly selects recent years. An alternative is the western pole **relative to the tropical mean** (the logic RONI applies to the Pacific):

```
wio_anom = ds.Index.custom(name="wio_anom", regions={"west": [-10, 10, 50, 70]},
                           combine=lambda b: b["west"], transform="anomaly",
                           weights="cos_lat", baseline=BASELINE).reduce(ond)
trop      = ds.Index.custom(name="trop", regions={"trop": [-20, 20, 0, 360]},
                           combine=lambda b: b["trop"], transform="anomaly",
                           weights="cos_lat", baseline=BASELINE).reduce(ond)
wio_rel   = wio_anom - trop      # western pole warmth ABOVE the tropical-mean warming

comp = pd.DataFrame({"WIO abs (°C)": wio.to_series().round(2),
                    "WIO rel-to-tropics (°C)": wio_rel.to_series().round(2),
                    "IOD (°C)": iod.to_series().round(2),
                    "EEA rain": pd.Series({int(y): str(c) for y, c in zip(rain.year.values,
rain_tercile.values)}}))
comp.loc[[1997, 2015, 2019, 2023]]
```

	WIO ABS (°C)	WIO REL-TO-TROPICS (°C)	IOD (°C)	EEA RAIN
1997	28.639999	-0.03	1.14	Above
2015	29.120001	0.23	0.35	Above
2019	28.920000	0.37	1.23	Above
2023	29.190001	0.25	1.25	Above

On the relative metric 2015 is unremarkable — its warm western pole was largely basin-wide warming rather than a local anomaly, consistent with its weak dipole. 1997, 2019 and 2023 remain strong. The dipole-plus-relative-warmth view therefore points to 1997/2019/2023 as the closer analogs.

6d · June-initialised forecast skill by index

The outlook rests on June-initialised NMME forecasts of the OND ocean state. A June start sits just before the Indian Ocean's boreal-summer IOD growth phase — the predictability barrier, across which dynamical OND-IOD skill is typically lower than OND-ENSO skill. Scoring the bias-corrected forecast against observations over the hindcast:

```
hind = list(range(BASELINE[0], BASELINE[1] + 1))
for name, fc, obs in [("RONI", roni_fc, roni), ("IOD", iod_fc, iod), ("WIO", wio_fc, wio)]:
    r = float(np.corrcoef(fc.sel(year=hind).values, obs.sel(year=hind).values)[0, 1])
    print(f" June-init OND {name:4s} hindcast skill (corr, 1991–2020): r = {r:+.2f}")
```

```
June-init OND RONI hindcast skill (corr, 1991–2020): r = +0.84
June-init OND IOD hindcast skill (corr, 1991–2020): r = +0.75
June-init OND WIO hindcast skill (corr, 1991–2020): r = +0.81
```

The IOD's June-initialised skill is the lowest of the three, so the strong-positive IOD component of the outlook carries the most forecast uncertainty and would firm up from an August/September initialisation.

Recap

With the rainfall colouring (§2b) and the forecast-vs-forecast axes (§4, slide 2), the notebook reproduces the deck's argument: forecast strong-El Niño-plus- strong-dipole years are the wet ones, and OND 2026 forecasts into that corner, nearest 1997 and 2023. The four notes in §6 — clamped vs extrapolated mapping, in-sample vs cross-validated bounds, absolute vs trend-relative WIO, and by-index forecast skill — do not change that direction, but bear on how much uncertainty to attach to it.